

Frank Liu

U.S. Citizen | Atlanta, GA | 404-490-6712 | fliu86725@gmail.com | frankliuuu.github.io | github.com/frankliuuu

EDUCATION

Princeton University

Princeton, NJ

Bachelor of Science in Engineering (B.S.E) in Computer Science

Sept. 2022 – May 2026

- **Relevant Coursework:** Algorithms & Data Structures, Machine Learning, Linear Algebra, Data Science, Building a Startup, Computer System Design

Georgia Institute of Technology

Atlanta, GA

Dual Enrollment

Aug. 2021 – May 2022

EXPERIENCE

Software Development Engineer Intern

May 2025 – August 2025

Amazon Web Services, Kiro Team

Seattle, WA

- Developed and shipped core onboarding and migration features for Kiro, AWS's agentic AI IDE, as part of the launch team
- Automated repository configuration imports from Cursor, Windsurf, and Cline using TypeScript, React, and Node.js, reducing developer setup friction
- Built and tested VS Code extension features with Mocha, Chai, and Sinon using test-driven development, improving reliability and maintaining high test coverage

Undergraduate Teaching Assistant

Jan. 2024 – May 2026

Princeton University

Princeton, NJ

- Lead weekly lab sessions and provide technical support for 400+ students studying data structures, algorithms, and systems programming
- Debug and optimize Java and C implementations, helping students understand complex data structures, algorithms, and systems concepts

Product Design Intern

Dec. 2023 – Jan. 2024

22x36 - Chinese Language Learning App

Remote

- Researched rewards and progress-tracking gamification strategies to improve user engagement with interactive practice features
- Leveraged generative AI tools including ChatGPT, DALL·E 3, and Claude to develop interactive HSK-aligned language-learning content and inform early product design

PROJECTS

TigerSpot | *Python, Flask, JavaScript, PostgreSQL*

- Built and deployed a full-stack campus location-guessing game using Flask, JavaScript, and PostgreSQL, serving 200+ Princeton users
- Designed a relational PostgreSQL schema for player profiles, rankings, leaderboards, and points, and implemented asynchronous multiplayer gameplay
- Deployed backend infrastructure with Render and optimized image delivery/storage through Cloudinary

Digital Wardrobe App | *Swift/SwiftUI*

- Built a SwiftUI mobile application for uploading, categorizing, and organizing clothing items into customizable outfit collections through Spec Driven Development

Word Hunt Solver and Learner | *React, Flask, Python*

- Engineered a Word Hunt solver using Trie-based prefix pruning and recursive DFS/backtracking
- Built a full-stack web application with React and Flask REST APIs, implementing interactive 4×4 board input, solution rendering, path visualization, and dictionary definitions

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, C/C++, HTML/CSS, SQL, R, Swift

Technologies: React, Node.js, Flask, PostgreSQL, SQLite, SwiftUI, NumPy

Tools: Git, Kiro, Mocha, Chai, Sinon, Claude